

Eqn. (3) is the fundamental equation governing the steady-state handling behavior of a road vehicle (with a rigid frame).

Note:

a) The steer angle required to negotiate a given curve depends on:

- i) the wheelbase, L
- ii) weight distribution (L_1 & L_2)
- iii) forward speed, V
- iv) tire cornering stiffnesses, $C_{\alpha f}$ & $C_{\alpha r}$

b) Alternative expression for $K_{us} = W \frac{(L_2 C_{\alpha r} - L_1 C_{\alpha f})}{L C_{\alpha f} C_{\alpha r}}$